Self-Supervised Learning

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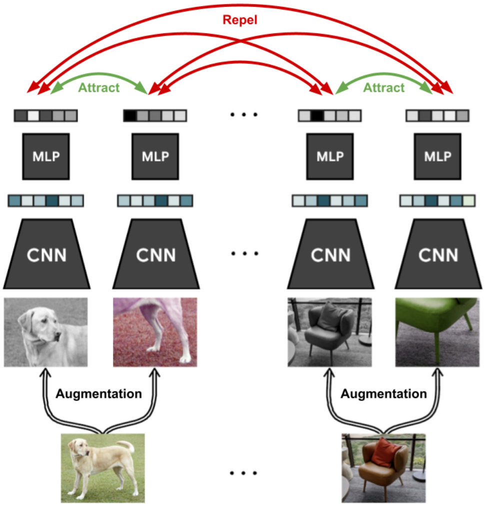
July 3, 2026


Repel
Attract
Attract
MLP
MLP
MLP
MLP
CNN
CNN
CNN
CNN
Augmentation
Augmentation

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Motivation for Self-Supervised Learning

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   Annotating data is expensive and slow, but unlabeled data is everywhere!

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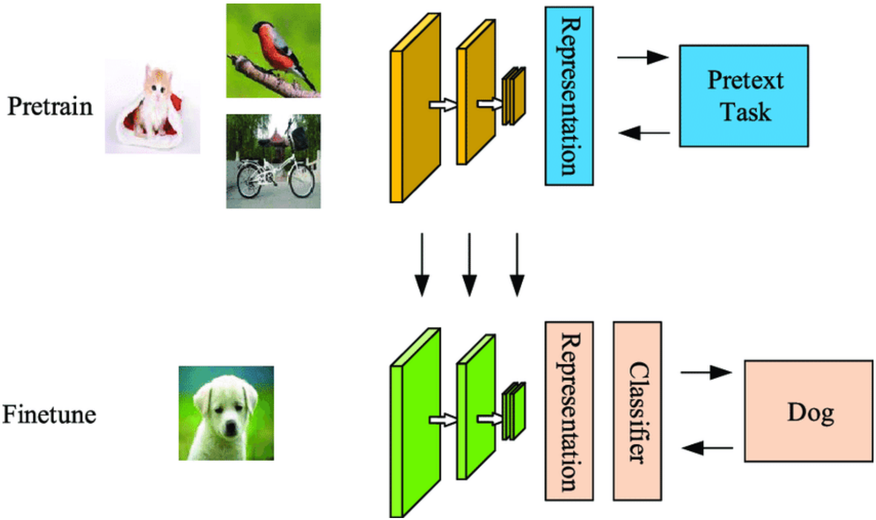
5. **A journey of methods:**
   The field has evolved from generative models (like VAEs and GANs), to contrastive and predictive coding, and now to hybrid approaches combining reconstruction and discrimination.


Pretrain
Representation
Pretext Task
Finetune
Representation
Classifier
Dog

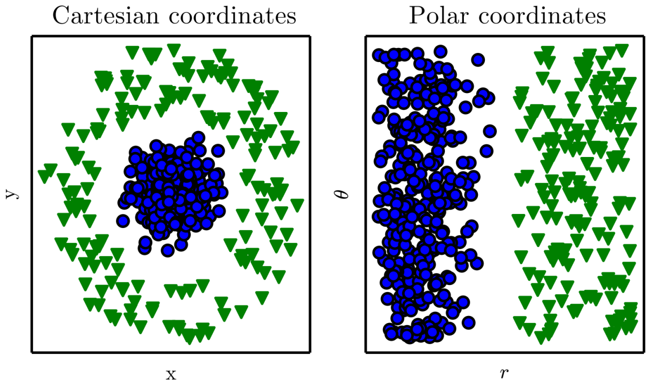
SSL: Learning Outcomes

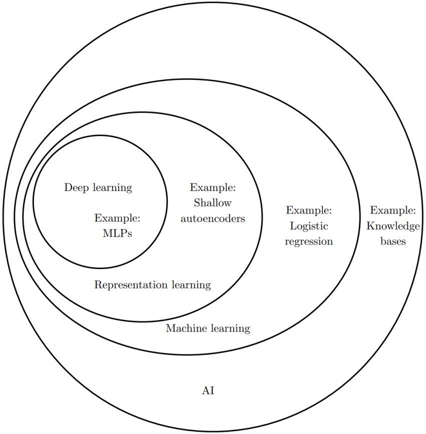
By the end of this session, you should be able to:

- **Explain** the data efficiency imperative and theoretical motivations behind self-supervised learning (SSL).
- **Identify** key cognitive principles that inspire self-supervision in machine learning.
- **Describe** Instance Discrimination and **implement** the Momentum Contrast (MoCo) framework.
- **Critically assess** pretext tasks such as rotation prediction, relative patch prediction, and view prediction.

SSL: **Introduction**

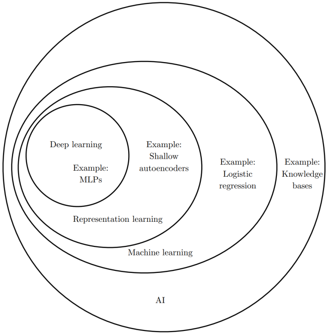
Representations Matter





Deep Unsupervised Learning:

- Learns data representations without using labels.
- Is a subset of Deep Learning, which itself is a subset of Representation Learning, all within Machine Learning.



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Self-Supervised Learning (SSL):

- A form of unsupervised learning that creates supervision from the data itself by designing *pretext tasks*.
- These tasks generate pseudo-labels, enabling the model to learn useful representations without manual annotation.
- Often used interchangeably with unsupervised learning, but SSL specifically focuses on leveraging intrinsic data structure.

Why Self-Supervised Learning?

- **High cost of dataset creation:** Each new task often requires building a new labeled dataset.
  - Involves preparing labeling manuals, defining categories, hiring annotators, building GUIs, and managing storage pipelines.
- **Expensive supervision:** High-quality labels can be costly or impractical to obtain (e.g., in medicine or legal domains).
- **Leverage unlabeled data:** The internet provides vast amounts of unlabeled images, videos, and text that can be utilized.
- **Cognitive motivation:** Animals and babies learn from their environment without explicit supervision, inspiring SSL approaches.

"Give a robot a label and you feed it for a moment; teach a robot to label and you feed it for a lifetime."
— Pierre Sermanet

This quote highlights the core motivation behind self-supervised learning
→ enabling machines to generate their own supervision from data
→ thus reducing reliance on costly manual labeling and
→ fostering scalable, autonomous learning.

What is Self-Supervised Learning?

SSL is a type of unsupervised learning where the data itself provides supervision. Typically, part of the data is hidden, and a neural network is trained to predict it from the rest—this defines the *pretext task*. Well-chosen pretext tasks help models learn useful features without manual labels.

- **Pretext Task**: Examples include predicting masked patches, future representations (CPC), or distinguishing positive/negative pairs (contrastive).
- **Downstream Task**: After pre-training, the encoder is fine-tuned for tasks like classification or detection.
- **Key Trade-Offs**:
  - **Reconstructive** (e.g., autoencoders): capture low-level details, may miss semantics.
  - **Predictive/Contrastive** (e.g., CPC, MoCo): focus on high-level information.

What is Self-Supervised Learning? (cont.)

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- **"Pure" Reinforcement Learning (cherry)**
  - The machine predicts a scalar reward given once in a while.
  - **A few bits for some samples**
- **Supervised Learning (icing)**
  - The machine predicts a category or a few numbers for each input
  - Predicting human-supplied data
  - **10→10,000 bits per sample**
- **Self-Supervised Learning (cake génoise)**
  - The machine predicts any part of its input for any observed part.
  - Predicts future frames in videos
  - **Millions of bits per sample**

SSL: **Cognitive Principles**

Cognitive Principles

- **Predictive Coding**: The brain optimizes to predict future sensory inputs.
- **Error Correction**: Learning signals arise from reconstructive or contrastive errors.
  - *Reconstruction from Corrupted Inputs*:
    - Denoising autoencoders
    - Inpainting
    - Colorization, split-brain autoencoders
  - *Visual Common Sense Tasks*:
    - Relative patch prediction
    - Jigsaw puzzles
    - Rotation prediction
  - *Contrastive Learning*:
    - word2vec

Cognitive Principles (cont.)

- Contrastive Predictive Coding (CPC)
- Instance discrimination
- Recent state-of-the-art methods

- **Multi-View Perception**: Integrating different sensory views enhances representations.

SSL: **Pretext Task Taxonomy**

Pretext Task Taxonomy: Introduction

Pretext tasks are broadly categorized by their supervision signal and learning objective:

- **Reconstructive Methods**: Focus on reconstructing parts or properties of the input data.
- **Predictive Methods**: Involve predicting withheld or transformed aspects of the data.

Each category includes various techniques that have advanced SSL in computer vision.

Understanding this taxonomy helps in:

- Selecting suitable pretext tasks for specific applications.
- Gaining insight into how SSL methods learn robust and transferable representations.

SSL: **Instance Discrimination & Contrastive Learning**

Instance Discrimination & Contrastive Learning

Pretext Task Evolution:

- Earlier: handcrafted tasks (e.g., jigsaw, colorization).
- Now: contrastive learning directly optimizes for invariance and discrimination.
- Leads to better, more transferable features and narrows the gap with supervised learning.

Instance Discrimination:

- Treats each image as its own class.
- Positive pairs: augmented views of the same image.
- Negative pairs: views from different images.
- Promotes invariance to augmentations and discrimination between instances.

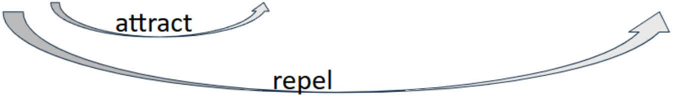
Contrastive Learning:

- Self-supervised approach for learning representations, especially in vision.
- Uses data structure instead of labels.
- Learns by pulling together similar (positive) pairs and pushing apart dissimilar (negative) pairs using a contrastive loss (e.g., InfoNCE).

Fundamentals of Instance Discrimination

Instance Discrimination is a self-supervised learning approach where each individual sample in the dataset is treated as a distinct class.

- The core idea is to learn representations by distinguishing between different instances, rather than relying on human-annotated labels.
- **Positive pairs** are generated by applying different augmentations (such as cropping, color jittering, or flipping) to the same image. These augmented views are considered to belong to the same class (i.e., the same instance).
- **Negative pairs** are formed by pairing augmented views of different images. These are treated as belonging to different classes (i.e., different instances).



1. MoCo
2. SimCLR

Fundamentals of Instance Discrimination (cont.)

- The learning objective is to **maximize agreement** (similarity) between representations of positive pairs, while minimizing agreement between negative pairs.
- This is typically achieved using a contrastive loss function, such as InfoNCE, which encourages the model to bring positive pairs closer in the embedding space and push negative pairs apart.
- Instance discrimination forms the basis for many popular self-supervised learning methods, such as MoCo, SimCLR, and PIRL.
- By maximizing agreement across views of the same instance, the model learns to extract meaningful and invariant features, which can be transferred to downstream tasks.

Momentum Contrast for Unsupervised Visual Representation Learning

Kaiming He Haoqi Fan Yuxin Wu Saining Xie Ross Girshick

Facebook AI Research (FAIR)

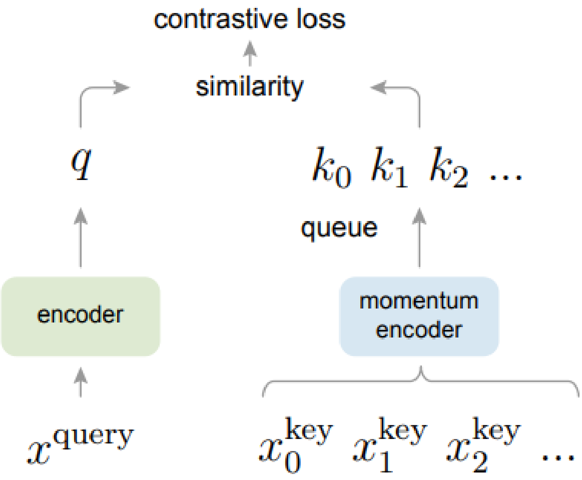
Momentum Contrast (MoCo) (cont.)

Momentum Contrast (MoCo) is a self-supervised learning framework designed for visual representation learning. Its key components are:

Query & Key Encoders: Two neural networks, f_q (query encoder) and f_k (key encoder), are used. The key encoder f_k is updated as an exponential moving average of the query encoder f_q:

$$\theta_k \leftarrow m\theta_k + (1 - m)\theta_q$$

where θ_k and θ_q are the parameters of f_k and f_q, and m is the momentum coefficient (e.g., $m = 0.999$).


contrastive loss
similarity
q
$k_0 \; k_1 \; k_2 \; \ldots$
queue
encoder
momentum encoder
x^{query}
$x_0^{\text{key}} \; x_1^{\text{key}} \; x_2^{\text{key}} \; \ldots$

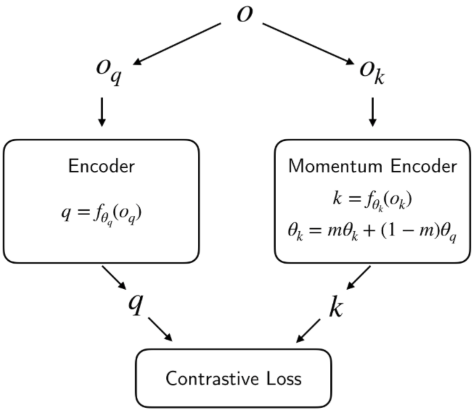
Dictionary Queue: MoCo maintains a large queue (dictionary) of encoded keys from previous batches. This enables the use of a large and consistent set of negative samples for contrastive learning, which is crucial for effective representation learning.

Contrastive Loss: The InfoNCE loss is used to train the encoders. For a given query q and its positive key k^+, along with a set of negative keys $\{k_0, k_1, ..., k_N\}$ from the dictionary, the loss is:

$$\mathcal{L}_q = -\log \frac{\exp(q \cdot k^+/\tau)}{\exp(q \cdot k^+/\tau) + \sum_{i=0}^{N} \exp(q \cdot k_i/\tau)}$$

where τ is a temperature hyperparameter.

Momentum Contrast (MoCo) (cont.)



$$\mathcal{L}_q = -\log \frac{\exp(q \cdot k_+ / \tau)}{\sum_{i=0}^{K} \exp(q \cdot k_i / \tau)}$$

Key Steps in MoCo Training:

1. For each image, generate two augmentations: one for the query encoder (f_q), one for the key encoder (f_k).
2. Encode the query and key.
3. Compute the InfoNCE loss using the current positive key and the dictionary of negative keys.
4. Update f_q via backpropagation; update f_k via momentum.
5. Enqueue the new key and dequeue the oldest key to maintain the dictionary size.

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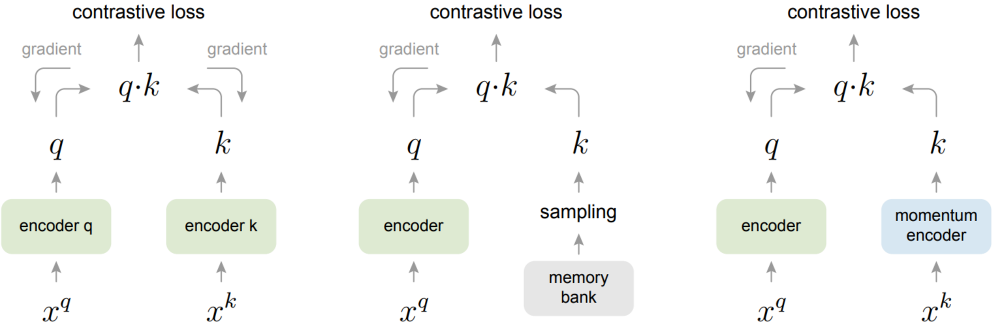
Algorithm 1 Pseudocode of MoCo in a PyTorch-like style.

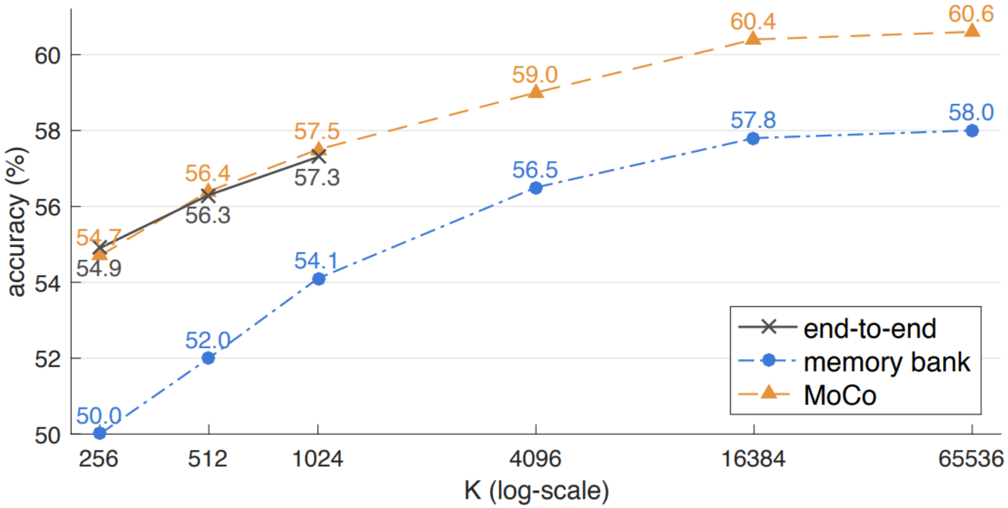
```
# f_q, f_k: encoder networks for query and key
# queue: dictionary as a queue of K keys (CxK)
# m: momentum
# t: temperature

f_k.params = f_q.params # initialize
for x in loader: # load a minibatch x with N samples
   x_q = aug(x) # a randomly augmented version
   x_k = aug(x) # another randomly augmented version

   q = f_q.forward(x_q) # queries: NxC
   k = f_k.forward(x_k) # keys: NxC
   k = k.detach() # no gradient to keys

   # positive logits: Nx1
   l_pos = bmm(q.view(N,1,C), k.view(N,C,1))

   # negative logits: NxK
   l_neg = mm(q.view(N,C), queue.view(C,K))

   # logits: Nx(1+K)
   logits = cat([l_pos, l_neg], dim=1)

   # contrastive loss, Eqn.(1)
   labels = zeros(N) # positives are the 0-th
   loss = CrossEntropyLoss(logits/t, labels)

   # SGD update: query network
   loss.backward()
   update(f_q.params)

   # momentum update: key network
   f_k.params = m*f_k.params+(1-m)*f_q.params

   # update dictionary
   enqueue(queue, k) # enqueue the current minibatch
   dequeue(queue) # dequeue the earliest minibatch
```

bmm: batch matrix multiplication; mm: matrix multiplication; cat: concatenation.

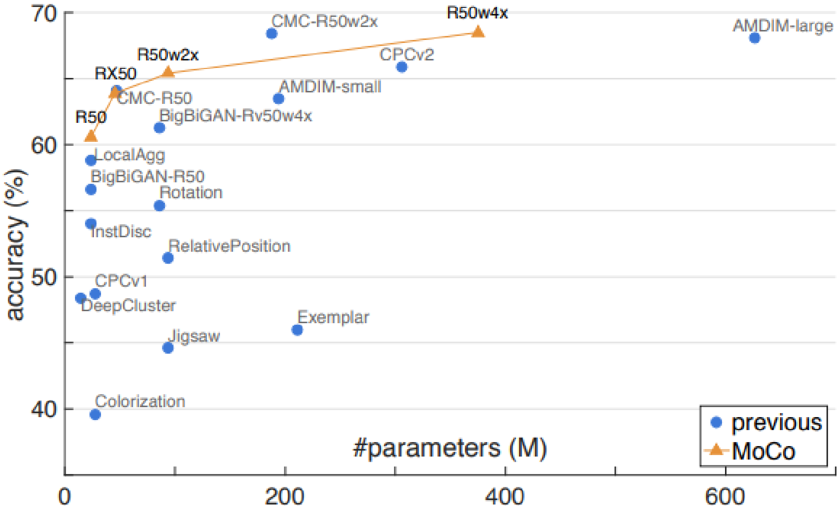


(a) end-to-end (b) memory bank (c) MoCo

Momentum Contrast (MoCo) (cont.)



Momentum Contrast (MoCo) (cont.)



Key Advantages:

- Enables large and consistent negative sets for contrastive learning.
- Momentum update stabilizes the key encoder, improving training.
- Scalable to large datasets and high-dimensional representations.

SSL: **References**

References

[1] Vincent, P., Larochelle, H., Bengio, Y., & Manzagol, P.-A. (2008). *Extracting and Composing Robust Features with Denoising Autoencoders*. ICML.

[2] Pathak, D., Krahenbuhl, P., Donahue, J., Darrell, T., & Efros, A. A. (2016). *Context Encoders: Feature Learning by Inpainting*. CVPR.

[3] Noroozi, M., & Favaro, P. (2016). *Unsupervised Learning of Visual Representations by Solving Jigsaw Puzzles*. ECCV.

[4] Gidaris, S., Singh, P., & Komodakis, N. (2018). *Unsupervised Representation Learning by Predicting Image Rotations*. ICLR.

[5] van den Oord, A., Li, Y., & Vinyals, O. (2018). *Representation Learning with Contrastive Predictive Coding*. arXiv:1807.03748.

[6] Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). *Efficient Estimation of Word Representations in Vector Space*. arXiv:1301.3781.

[7] He, K., Fan, H., Wu, Y., Xie, S., & Girshick, R. (2020). *Momentum Contrast for Unsupervised Visual Representation Learning*. CVPR.